

AP CALCULUS AB – UNIT 2

2.1 Rates of Change and Tangent Lines



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Calculate average rate of change from formulas, tables, graphs, and contexts.
- Interpret secant-line slope and tangent-line slope with correct units.
- Use shrinking intervals to motivate and estimate instantaneous rate of change.

Keys: Essential Knowledge and Vocabulary

Average rate of change	$\frac{f(b) - f(a)}{b - a}$, the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.
Instantaneous rate of change	The limiting value of average rates as the interval collapses to a point; geometrically, the tangent-line slope.
Difference quotient	$\frac{f(a + h) - f(a)}{h}$ for $h \neq 0$.
Units	Output units divided by input units; for position in meters and time in seconds, the rate is meters per second.

Explanation, Strategy, and Common Errors

- An average rate describes net change across an interval. It does not claim that the rate stayed constant inside the interval.
- To approximate a rate at $x = a$, use secant slopes over intervals with endpoints increasingly close to a . A symmetric interval often reduces one-sided bias.
- In contextual answers, state the quantity, sign, units, and time or input value. A negative rate means the dependent variable is decreasing with respect to the independent variable.

Solved Example: Secant slope from a formula

For $s(t) = t^3 - 4t$ meters, find the average velocity on $[1, 3]$.

Answer and Reasoning

$$s(3) = 15 \text{ and } s(1) = -3, \text{ so } \frac{15 - (-3)}{3 - 1} = 9 \text{ meters per second.}$$

Solved Example: A shrinking interval

For $f(x) = x^2 + 1$, calculate the average rate on $[2, 2 + h]$ and identify its limiting value as $h \rightarrow 0$.

Answer and Reasoning

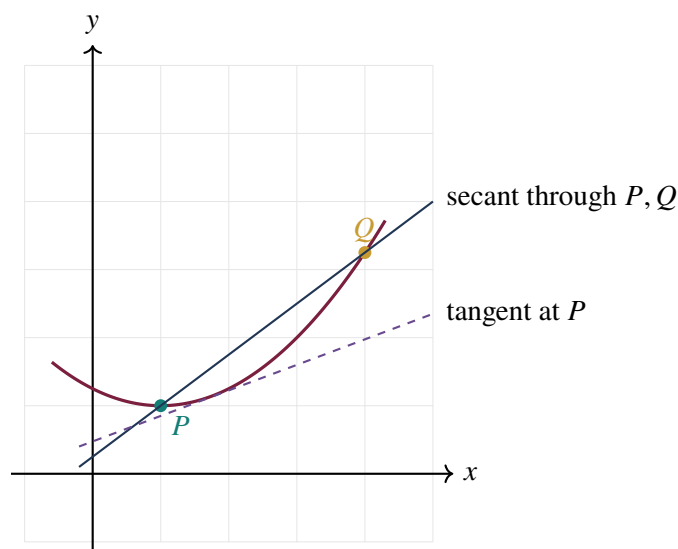
$$\frac{(2+h)^2 + 1 - 5}{h} = \frac{4h + h^2}{h} = 4 + h. \text{ As } h \rightarrow 0, \text{ the rate approaches } 4.$$

Solved Example: Interpretation with units

The temperature $T(t)$ of a reaction is measured in $^{\circ}\text{C}$, where t is in minutes. Explain the meaning of $\frac{T(8)-T(5)}{3} = -1.6$.

Answer and Reasoning

From minute 5 to minute 8, the temperature decreased at an average rate of 1.6°C per minute.

Shared Representation / Data**Rates of Change****AP Exam Language**

For every derivative in context, include the sign, units, input value, and a complete verbal interpretation. For every existence claim, state the relevant hypothesis or one-sided comparison.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

- Find the average rate of change of $f(x) = 3x^2 - 2$ on $[1, 4]$.

- Find the average rate of change of $g(x) = \sqrt{x+1}$ on $[0, 8]$.

3. A population changes from 820 to 910 organisms over 6 hours. Find and interpret the average rate.

4. Write a difference quotient that estimates the instantaneous rate of p at $x = 7$ using an increment h .

5. For $f(x) = 5 - 2x$, compare the average rate on $[-3, 1]$ with the instantaneous rate at any point.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

6. For $s(t) = t^3 - 6t^2 + 9t$, find the average velocity on $[1, 2]$ and on $[1, 1.5]$.

7. Use $f(x) = \frac{1}{x}$ to find the secant slope through $x = 2$ and $x = 2 + h$. Simplify.

8. A tank contains $V(t) = 1200 - 18t - 0.6t^2$ liters. Find the average rate of change on $[4, 7]$ and interpret the sign.

9. Given $f(2.9) = 7.41$, $f(3) = 7.70$, and $f(3.1) = 8.01$, estimate $f'(3)$ using a symmetric secant.

10. Explain why $\frac{f(a)-f(a)}{a-a}$ cannot define an instantaneous rate directly.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

11. For $f(x) = x^2 - 5x$, determine all intervals $[a, a + 2]$ on which the average rate is 3.

12. If the average rate of f on every interval $[a, b]$ equals 7, what form must f have? Justify.

13. A particle has $s(t) = t^4 - 8t^2$. Find the average velocity on $[2 - h, 2 + h]$ and its limit as $h \rightarrow 0$.

14. Rates over $[4, 4.2]$, $[4, 4.02]$, and $[4, 4.002]$ are -3.18 , -3.018 , and -3.0018 . Estimate the instantaneous rate and explain.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

15. For $f(x) = \frac{x-1}{x+2}$, derive the average rate on $[a, a + h]$ in simplified form.

16. Find all c such that the average rate of $f(x) = x^3$ on $[c - 1, c + 1]$ equals the average rate on $[-1, 1]$.

17. A differentiable function has $f(5) = 12$. Secant slopes through $(5, 12)$ and $(5 + h, f(5 + h))$ are always greater than 4 for $0 < h < 0.1$ and approach a finite limit. What can be concluded about the right-hand instantaneous rate?

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

18. For $f(x) = |x|$, compute the average rates on $[-h, 0]$ and $[0, h]$ for $h > 0$. What do they predict at 0?

19. Suppose $\frac{f(a+h)-f(a)}{h} = 2a + 3 + h \sin(1/h)$ for $h \neq 0$. Determine the instantaneous rate at a .

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. The average rate of change of f on $[a, b]$ is

- (A) $\frac{f(a)-f(b)}{b-a}$ (B) $\frac{f(b)-f(a)}{b-a}$
(C) $f(b) - f(a)$ (D) $\frac{b-a}{f(b)-f(a)}$

2. For $f(x) = x^2$, the average rate on $[3, 3 + h]$ is

- (A) 6 (B) $6 + h$
(C) $9 + h^2$ (D) $3 + h$

3. A car position is measured in kilometers and time in hours. The units of an instantaneous rate are

- (A) hours per kilometer (B) kilometers
(C) kilometers per hour (D) hours

4. Which statement is always true?

- (A) An average rate equals the instantaneous rate at every point. (B) A negative average rate means the function is negative.
(C) An instantaneous rate is modeled by a limit of secant slopes. (D) Secant slopes exist even when the endpoints have the same input.

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. A calculator gives $f(1.999) = 6.994001$ and $f(2.001) = 7.006001$. A symmetric estimate of $f'(2)$ is

- (A) 3 (B) 6
(C) 7 (D) 12

6. Water volume changes from 431.8 L at $t = 2.5$ to 418.1 L at $t = 3.2$. The average rate is closest to

- (A) -19.57 L/min (B) -9.59 L/min
(C) 9.59 L/min (D) 19.57 L/min

AP-Style Free-Response Practice

Calculator Permitted

Complete the following independent calculator-permitted problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Rates of Change. A particle moves on a line with position $s(t) = t^3 - 5t^2 + 4t$, measured in meters. Find the average velocity on $[1, 3]$. [2 points]

(1b) Write and simplify the difference quotient based at $t = 2$. [3 points]

(1c) Use part (b) to determine the instantaneous velocity at $t = 2$ and interpret it. [2 points]

No Calculator

Complete the following independent non-calculator problems. Show the mathematical setup and justify conclusions when requested.

(1a) Problem 1: Rates of Change. The concentration $C(t)$, in milligrams per liter, is sampled near $t = 6$ minutes. Using $C(5.98) = 14.822$, $C(6.00) = 14.760$, and $C(6.02) = 14.696$, estimate $C'(6)$ with a symmetric difference. [2 points]

(1b) Explain the meaning of your estimate in context. [2 points]

(1c) The one-sided estimates are -3.10 and -3.20 . Explain why the symmetric estimate is reasonable. [2 points]

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